

PROJECT TITLE : AI FAQ ASSISTANT API

PROJECT PLANNING PHASE

1. PROJECT OBJECTIVE

The main objective of the AI FAQ Assistant is to develop a secure backend REST API for creating, managing, searching, and generating FAQs using AI. The project focuses on JWT authentication, FAQ CRUD operations, category management, keyword-based search, secure data storage, and AI-assisted FAQ generation.

2. TECHNOLOGY PLANNING

The project uses Node.js and Express.js for backend development, MongoDB and Mongoose for database management, and Google Gemini AI for generating FAQ questions and answers. JWT, bcrypt, dotenv, and CORS are used for authentication, password security, configuration, and API communication. Postman or Thunder Client is planned for API testing.

3. MODULE PLANNING

The system is planned with modules for user authentication, FAQ management, category management, FAQ search, AI FAQ generation, database operations, and error handling. The backend includes authentication routes, FAQ CRUD routes, AI generation routes, authentication middleware, and centralized error handling.

4. DATABASE PLANNING

MongoDB is selected as the primary database, with Mongoose used for schema definition and validation. The main entities include User, FAQ, AI Generation, and Category. The User and FAQ models store required information such as user details, roles, questions, answers, categories, creators, and timestamps.

5. SECURITY PLANNING

Security is planned using JWT-based authentication and bcrypt for password encryption. Different roles are provided with different access permissions. Private APIs are protected using JWT Bearer-token authentication to ensure secure access to the system.

6. TESTING AND EXECUTION PLANNING

The application will be tested using Postman or Thunder Client. Testing includes user registration, login, JWT token generation, protected API access, FAQ creation and management, FAQ search, and AI-based FAQ generation.

7. EXPECTED OUTCOME

The planned system will provide a secure backend application for authentication, FAQ management, searching, category organization, and AI-generated FAQ content.

